

Circuit Overview

Components

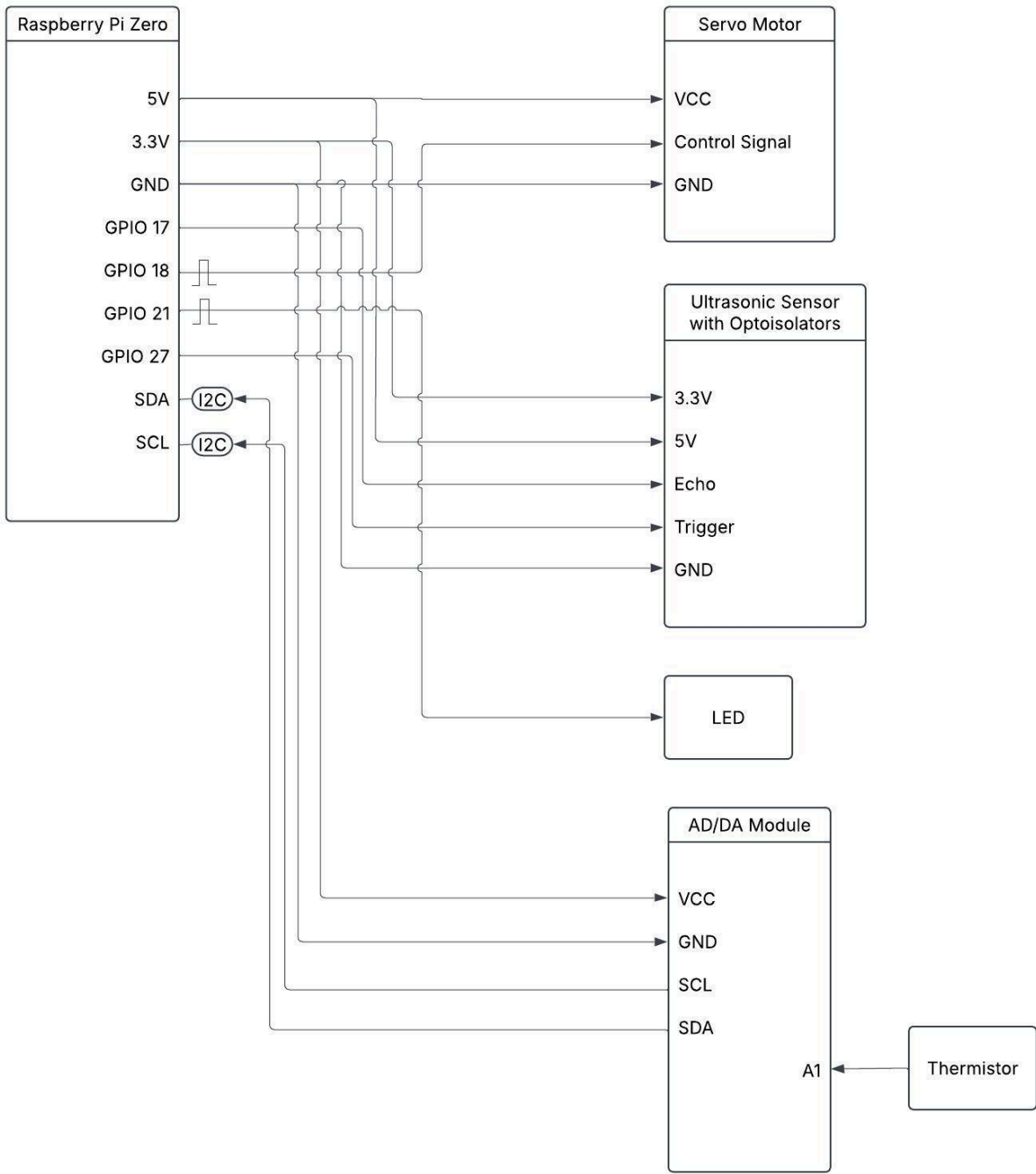
1. Ultrasonic Sensor
2. Thermistor on ADC/DAC board
3. LED
4. Servo Motor
5. Raspberry Pi Zero

The Raspberry Pi reads the analog temperature data of the room from the thermistor on the ADC/DAC board via I2C, and distance from the ultrasonic sensor. Based on these inputs, it controls the servo motor which simulates a fan and the LED's brightness, which is proportional to the temperature of the room.

Possible Scenarios

1. Room is cool
When the room is cool, the servo motor is idle as there is no need for the fan to be on. The LED is either off or very dim.
2. Room is warm
When the room is warm, the servo motor begins movement from left to right to simulate the fan, and moves at a slow speed. If the ultrasonic sensor detects the presence of a person, the fan speed increases to maximum speed. The LED is at medium brightness.
3. Room is hot.
When the room is hot, the fan speed is at maximum, with or without a person in the room. The LED is bright.

Block Diagram



Block Templates

Ultrasonic Sensor Block Template

Device name: Ultrasonic Ranging Module

Model number: HC - SR04

Supply voltage range: 5V

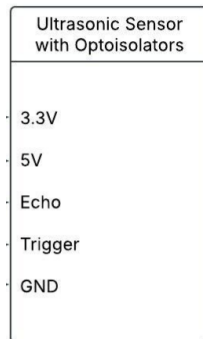
Input type(s): Digital 10uS TTL pulse

Output type(s): Digital TTL pulse proportionate to distance

Interface: Digital

Description: Measures distances ranging from 2cm to 4m.

Symbol:



Servo Motor Block Template

Device name: Servo Motor

Model number: SG90

Supply voltage range: 4.8 V (~5V)

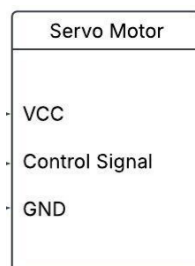
Input type(s): PWM, 50Hz, duty cycle $\leq 20\%$

Output type(s): N/A

Interface: N/A

Description: Servo motor that can rotate approximately 180 degrees (90 in each direction).

Symbol:



Thermistor via AD/DA Module Block Template

Device name: Thermistor

Model number: PCF8591

Supply voltage range: 3.3 V

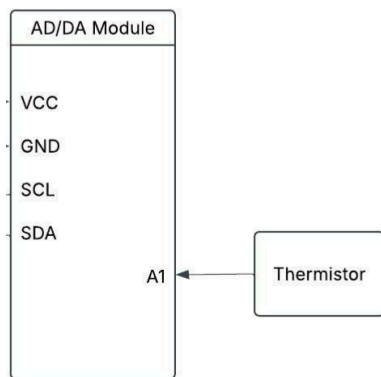
Input type(s): Analog

Output type(s): Analog

Interface: I2C

Description: Reads ambient temperature, outputs a voltage inversely proportional to temperature

Symbol:



Raspberry Pi Zero Block Template

Device name: Raspberry Pi Zero

Model number: W/WH

Supply voltage range: 5V

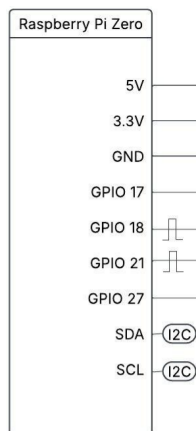
Input type(s): SPI / UART / I2C

Output type(s): PWM / I2C

Interface: I2C

Description: Central controller

Symbol:



Block Explanations

Raspberry Pi

Controls the system by reading sensor data, processes logic and controls outputs

Thermistor

Detects temperature changes in the environment (via PCF8591)

PCF8591

Converts analog thermistor signal to digital for Pi to read via I2C

Ultrasonic Sensor

Detects if a person is nearby based on distance measurement

Servo Motor

Acts as the fan, moves slowly or rapidly depending on inputs

LED

Indicates temperature level visually using PWM brightness